COFFEE PACKAGING QUALITY INSPECTION REPORT

Real-Time AI Seal Inspection and Physical Leak Detection

1. Report Information

Report ID	RPT-20260821-25F4CE
Inspection Date	21 August 2026
Inspection Time	08:51:01 PM
Inspection Module	Packet Seal and Leak Quality Inspection

2. Overall Inspection Summary

Inspection	Result
Real-Time AI Overheat Inspection	STOPPED
Physical Packet Leak Test	GOOD
Overall Quality Decision	REJECT

3. Final Quality Decision

REJECT

Potential overheat defect was detected by the real-time AI inspection system.

4. Real-Time Two-Stage AI Inspection

Detected Seal Regions	2
Overheat Detected	YES
Highest Overheat Confidence	63.8%
AI Inspection Status	STOPPED

Seal-by-Seal Results

Seal	Seal Confidence	Overheat	Overheat Confidence
Seal 1	84.0%	NO	-
Seal 2	83.6%	NO	-

Real-Time Inspection Evidence


PACKET: OVERHEAT_DETECTED
SEAL 84.1% | OVERHEAT
SEAL 83.6% | OVERHEAT
over heat 63.8%
over heat 54.4%
Coffee
Net Weight
6g

5. Physical Packet Leak Detection

Leak Test Result	GOOD
Initial Value	299812
Average Value	280053.9
Threshold	286000
Minimum	278754
Maximum	280804
Range	2050
Reading Count	10
Final Device Status	TOP_REACHED

Load Cell Readings

Reading Number	Sensor Value
1	278754
2	279204
3	279683
4	279911
5	280086
6	280274
7	280609
8	280568
9	280646
10	280804

6. Recommended Action

- Hold or reject the affected packet.

- Inspect sealing temperature settings.

- Verify sealing dwell time and heater condition.

- Inspect subsequent packets from the same batch.

7. Inspection System Information

AI Stage 1	Seal Region Object Detection
AI Stage 2	Overheat Defect Object Detection

Camera	IP Webcam Real-Time Video
Physical Controller	Arduino Uno
Force Measurement	HX711 Load Cell System
Mechanical Test	Motorized Packet Pressure Mechanism
Safety / Position Control	Top and Bottom Limit Switches

This report was generated automatically using the coffee packet quality inspection system.